



Handspring's new Treo 600 is the sign of things to come. It's a full PDA with a high-speed processor, integrated cell phone, colour display and a digital camera

are harmful for the environment.

Nickel-metal hydride (NiMH): The main advantage of NiMH over NiCd is the increased energy density, giving up to 30 per cent more power from the same size. They also do not suffer the same extent of memory effect as NiCd cells, but have a lower cycle life. They are preferred for cell phones and laptops due to their increased energy density and the absence of harmful toxic metals.

Lithium ion (Li-ion): Today's star performer, the Li-ion cell is fast replacing other technologies, especially as its cost price continues to fall. Lithium's light weight and excellent electro-chemical potential lets Li-ion batteries provide high voltage and excellent capacities with reduced weight. This is crucial for cell phones, where the battery makes up most of the weight of the device. Current Li-ion technology provides twice the energy densities of NiCd cells. Li-ion batteries are also the easiest to maintain, since they have no memory-effect problem and the self-discharge is much less than other chemistries. Li-ion is not without its drawbacks though. Li-ion batteries do not last very long—typically a year or two before performance starts to rapidly deteriorate. And this happens even if the battery is not in use. Another drawback is that the pack must have a protection circuit to regulate voltage during charge and use. They feature in higher-end cell phones, laptops, digital cameras and other expensive electronics. **Lithium ion polymer (Li-polymer):** Li-polymer is a derivative of the standard Li-ion with huge potential. The difference here is the type of electrolyte used—a solid polymer electrolyte with a little electrolyte gel added to increase ion conductivity. This

means the battery is safer and more stable than the standard Li-ion ones, and more importantly, smaller. They can be flexible and incorporated into the device's protective casing, or even into your clothing, measuring as little as 1 mm in thickness.

Li-polymer batteries perform about as well as standard Li-ion batteries, with nearly the same capacities and no real cost advantage. The packaging and size are the promises that have led mobile phone manufacturers to adopt this technology—the packaging can be made of foil, like food wraps. In the future though, capacities should increase, as techniques to eliminate the gel electrolyte and maintain safe operating temperatures are being researched.

Fill 'er up

By far the most promising future power source for mobile devices is the fuel cell. Combustion engines use the pressure generated by igniting the fuel and convert it into mechanical energy, to move your car. Fuel cells do not burn fuel like a combustion engine. They convert hydrogen and oxygen into water, producing electricity and heat as a by-product. The process is silent, clean, and uses fuel very efficiently. The cells will continue to produce power, as long as hydrogen and oxygen are supplied.

How do they work?

Like batteries, fuel cells generate electricity through a chemical reaction. Hydrogen enters the cell at the anode where a chemical reaction separates the electrons. The positively charged hydrogen atoms travel through the electrolyte to the cathode. The electrons are carried through a wire to the

load (the device using the power), and back to the cathode to complete the circuit. Oxygen enters the cell through the cathode, and reacts with the positively charged hydrogen atoms and the electrons from the electrical circuit to form water. This water is drained from the cell. Some fuel cell designs incorporate extra equipment to extract hydrogen from commonly available fuels such as natural gas and methanol, for added convenience. These systems are far less polluting than the internal combustion engine.

The types of fuel cell differ basically in the type of electrolyte used. Each type has its own unique characteristics. For example, the efficient, high powered alkali fuel cells, which generated power for the Apollo spacecraft, used a solution of potassium hydroxide as an electrolyte. But these run at temperatures close to 200 degrees Celsius, and risk electrolyte leakage.

Proton Exchange Membrane (PEM) cells use a thin permeable membrane of polymer as the electrolyte. The technology was invented by General Electric in the 1960s. They are less efficient than alkali cells but work at much more reasonable temperatures, making them more suited to consumer applications.

Micro Fuel Cells (MFC) will be used in most portable devices of the future, and the first prototypes generated 10 milliwatts of power. The US Department of Defence has funded the research through the Defence Advance Research Projects Agency (DARPA), to feed the mission critical devices of tomorrow's soldier. These troops will be carrying around a lot of fancy power-hungry gizmos that are currently



Maintenance

Nickle-based batteries: Nickel based batteries (NiCd or NiMh) suffer from memory-effect build-up which reduces the performance of the battery. This memory effect has been greatly reduced with newer technologies, but still remains an issue. This is caused due to excessive charging, or repeated charging without discharge. To avoid the problem, do not leave batteries in the charger for more than a few hours after the charger indicates full charge. Also, periodically discharge the batteries completely. If you do this at regular intervals, the usage patterns between discharges become less important. Using higher quality chargers also improves the service life of your battery.

Lithium-based batteries: Lithium batteries

do not suffer from memory-effect like NiCd and NiMh batteries do. In fact, they last better if they are not fully discharged every time they are used. Some engineers claim that batteries with the ability to indicate the level of charge do suffer from 'digital memory'—a term used to refer to the inaccurate indication of the level of charge. This indication becomes increasingly inaccurate over time. To resolve this you can perform a full discharge about once a month to recalibrate this indicator. Lithium batteries degrade when exposed to heat, so avoid exposure to heat and operate the batteries in a cooler environment whenever possible. Do not buy spare batteries if you don't plan to use them for a while.